

DISSERTATION PROPOSAL · SEPTEMBER 2026

Teaching AI Physical Intuition

Integrating Neural Physics Engines
into Embodied Foundation Models
for Enhanced Physical Reasoning
and Robust Generalization

Genesis-driven Physical CoT · Sim-to-Real Transfer

THE PROBLEM

Today's embodied AI is a **black box.**

Current VLA models predict actions from statistical probability — with no understanding of physical laws. When they meet an unseen material or shape, they break.

WHY IT MATTERS

Zero-shot

generalization on unseen
deformable objects remains an
unsolved bottleneck in embodied
AI

Contact-rich tasks: garment folding, cable insertion, socket alignment

AGENDA

What this proposal covers.

01 Problem & Motivation

Why physical reasoning matters

02 Core Contributions

Three innovations

03 Methodology

Genesis · Physical CoT · Spatial RAG

04 Validation Plan

Benchmarks & sim-to-real

05 Systems Optimization

Inference acceleration

PART I

Core Contributions

Three Core Contributions

01

Physical Chain-of-Thought

A structured, causally-grounded reasoning process forcing the model to output physical understanding before action tokens — breaking the black-box decision paradigm.

02

Differentiable Physics Training

Backpropagating physical gradients from Genesis directly into the foundation model — replacing gradient-free RL with physics-grounded training.

03

Zero-shot Generalization

Demonstrating that physical understanding enables manipulation of unseen deformable objects — bridging the sim-to-real gap.

CONTRIBUTION 1 · PHYSICAL CoT

Before vs. After: Decision-Making

PAST

Black-Box VLA

- × Sees image → statistical guess → action
- × No explicit physical reasoning
- × Fails on unseen materials or shapes
- × Brittle, correlational pattern matching

OURS

Physical CoT

- ✓ Observation → Constraint → Causal link → Action
- ✓ Forces explicit physical parameter output
- ✓ Machine-parseable state trajectory
- ✓ Causal graphs prevent brittle failures

PART II

Methodology

ARCHITECTURE

Four integrated components.

01

Neural Physics Engine (Genesis)

Universal differentiable physics — rigid, fluid, and soft-body in one system. Physical gradients backpropagated directly to the control policy.

02

Causal Physical CoT

Structured reasoning following Observation → Constraint → Causal link → Action. Informed by PhysX-CoT and CausalPhys patterns.

03

Spatial RAG & Episodic Memory

Text-based RAG transformed into 3D scene graphs. Real-time reasoning over spatial geometry and historical physical interactions.

04

Heterogeneous Data Pipeline

Exploiting the Embodied Data Pyramid: real robot, UMI-style, egocentric video, simulation, and general-purpose data.

WHY GENESIS

The ideal validation platform.

DIFFERENTIABLE

Full-chain backpropagation

MPM/FEM solvers are differentiable end-to-end — physical gradients flow back to the neural network.

GPU-NATIVE

10M+ FPS on a single GPU

Python + CUDA/Taichi backend. No USD scene graph overhead.

SENSORS

Multi-modal simulation

Ray-traced rendering, contact forces, depth maps, 3D point clouds, and interaction trajectories.

SOFT-BODY

Deformable object support

Unified rigid, fluid, and soft-material physics — directly targets garment folding and cable tasks.

BENCHMARKS

Built-in reference scenarios

Franka manipulation, tactile sandbox, cloth simulation — ready for baseline reproduction.

STANDARD

De facto since 2024–2025

Developed across 20+ research labs. The emerging standard for embodied AI simulation.

PART III

Validation & Optimization

VALIDATION PLAN

Baseline vs. Outstanding completion.

BASELINE

Simulation-only

- × Reproduce existing benchmarks
- × Physical CoT on simple grasping
- × Genesis evaluation environment

OUTSTANDING

Novel systems contribution

- ✓ Deformable object manipulation
- ✓ Zero-shot generalization proof
- ✓ Sim-to-real transfer validation
- ✓ Inference optimization

Systems Contribution: Inference Optimization

01

Heterogeneous Scheduling

CPU/GPU hybrid pipeline overlapping base model inference with physics engine computation — inspired by KTransformers.

02

Spatial RAG Acceleration

CUDA-fused KNN/ANN operators and memory-hierarchy-optimized trajectory caching to keep real-time retrieval off the critical path.

03

Physical CoT Decoding

Speculative and constrained decoding for the fixed Observation → Constraint → Action structure — reducing TTFT and generation latency.

We're building a **bridge** between the open-source brain and the physical world — using differentiable physics and physical chain-of-thought to give AI real physical intuition.